# symembed: graded packing of simple cases, written out in full

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Each section shows the graded embedding (`embedder="bursar_graded"`, notes Section 3.4) of one case: the chosen $h$ (order $r$) and $s$, the invariants output directly, the packed coordinates $z_j$ (exact complex linear forms in the $d$'s and $e$'s, at power $x^j$, with their $h$-weights), and every output expanded in the $d$'s and $e$'s with exact coefficients. Expansions with more than 100 terms show only their term count; all appear in full in `simple_cases_graded_full.txt`. Generated by `symembed/explore/simple_cases_graded.py`; do not edit by hand.

## 1 $S(2)$ acting on $(a, b)$, $k = 2$

$|G| = 2$, $m = 4$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 4 (plain Bursarial: 11; quotient dimension 3).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{ab}$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 1 | $d_{aa} - d_{bb}$ | $2e_{ab}$ |

| output | expanded |
|---|---|
| $inv[0]$ | $d_{aa} + d_{bb}$ |
| $inv[1]$ | $2d_{ab}$ |
| $ReE_{2,0}$ | $-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + 4e_{ab}^2$ |
| $ImE_{2,0}$ | $-4d_{aa}e_{ab} + 4d_{bb}e_{ab}$ |

## 2 $S(3)$ acting on $(a, b, c)$, $k = 2$

$|G| = 6$, $m = 9$ slots; $h = (abc)$ (order $r = 3$), $s = (bc)$, mode x; real coefficients. Outputs: 24 (plain Bursarial: 174; quotient dimension 5).

Invariants output directly:

- inv[0] $= 2d_{aa} + 2d_{bb} + 2d_{cc}$
- inv[1] $= 2d_{ab} + 2d_{ac} + 2d_{bc}$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 2 | $\frac{d_{ab}}{2} + \frac{d_{ac}}{2} - d_{bc}$ | $\frac{\sqrt{3}(d_{ab} - d_{ac})}{2}$ |
| 1 | 0 | 0 | $2e_{ab} - 2e_{ac} + 2e_{bc}$ |
| 2 | 1 | $2d_{aa} - d_{bb} - d_{cc}$ | $\sqrt{3}\,(-d_{bb} + d_{cc})$ |
| 3 | 2 | $\frac{3e_{ab}}{2} + \frac{3e_{ac}}{2}$ | $\frac{\sqrt{3}(-e_{ab} + e_{ac} + 2e_{bc})}{2}$ |

| output | expanded |
|---|---|
| $inv[0]$ | $2d_{aa} + 2d_{bb} + 2d_{cc}$ |
| $inv[1]$ | $2d_{ab} + 2d_{ac} + 2d_{bc}$ |
| $ReE_{2,2}$ | $-6d_{aa}d_{ab} - 6d_{aa}d_{ac} + 12d_{aa}d_{bc} - 6d_{ab}d_{bb} + 12d_{ab}d_{cc} + 12d_{ac}d_{bb} - 6d_{ac}d_{cc} - 6d_{bb}d_{bc} - 6d_{bc}d_{cc} + 12e_{ab}^2 - 24e_{ab}e_{ac} + 24e_{ab}e_{bc} + 12e_{ac}^2 - 24e_{ac}e_{bc} + 12e_{bc}^2$ |
| $ReE_{2,5}$ | $-18d_{aa}e_{ab} - 18d_{aa}e_{ac} + 18d_{bb}e_{ab} - 18d_{bb}e_{bc} + 18d_{cc}e_{ac} + 18d_{cc}e_{bc}$ |
| $ReE_{3,0}$ | $-2d_{ab}^3 + 3d_{ab}^2 d_{ac} + 3d_{ab}^2 d_{bc} + 3d_{ab}d_{ac}^2 - 12d_{ab}d_{ac}d_{bc} + 3d_{ab}d_{bc}^2 - 2d_{ac}^3 + 3d_{ac}^2 d_{bc} + 3d_{ac}d_{bc}^2 - 2d_{bc}^3$ |
| $ReE_{3,3}$ | $-24\sqrt{3}d_{aa}d_{ab}e_{ab} + 24\sqrt{3}d_{aa}d_{ab}e_{ac} - 24\sqrt{3}d_{aa}d_{ab}e_{bc} + 24\sqrt{3}d_{aa}d_{ac}e_{ab} - 24\sqrt{3}d_{aa}d_{ac}e_{ac} + 24\sqrt{3}d_{aa}d_{ac}e_{bc} - 9d_{ab}^2 e_{ac} - 9d_{ab}^2 e_{bc} + 18d_{ab}d_{ac}e_{ab} + 18d_{ab}d_{ac}e_{ac} + 24\sqrt{3}d_{ab}d_{bb}e_{ab} - 24\sqrt{3}d_{ab}d_{bb}e_{ac} + 24\sqrt{3}d_{ab}d_{bb}e_{bc} - 18d_{ab}d_{bc}e_{ab} + 18d_{ab}d_{bc}e_{bc} - 9d_{ac}^2 e_{ab} + 9d_{ac}^2 e_{bc} - 18d_{ac}d_{bc}e_{ac} - 18d_{ac}d_{bc}e_{bc} - 24\sqrt{3}d_{ac}d_{cc}e_{ab} + 24\sqrt{3}d_{ac}d_{cc}e_{ac} - 24\sqrt{3}d_{ac}d_{cc}e_{bc} - 24\sqrt{3}d_{bb}d_{bc}e_{ab} + 24\sqrt{3}d_{bb}d_{bc}e_{ac} - 24\sqrt{3}d_{bb}d_{bc}e_{bc} + 9d_{bc}^2 e_{ab} + 9d_{bc}^2 e_{ac} + 24\sqrt{3}d_{bc}d_{cc}e_{ab} - 24\sqrt{3}d_{bc}d_{cc}e_{ac} + 24\sqrt{3}d_{bc}d_{cc}e_{bc}$ |
| $ReE_{3,6}$ | $16d_{aa}^3 - 24d_{aa}^2 d_{bb} - 24d_{aa}^2 d_{cc} - 24d_{aa}d_{bb}^2 + 96d_{aa}d_{bb}d_{cc} - 24d_{aa}d_{cc}^2 + 24\sqrt{3}d_{aa}e_{ab}^2 - 48\sqrt{3}d_{aa}e_{ab}e_{ac} - 24\sqrt{3}d_{aa}e_{ab}e_{bc} + 24\sqrt{3}d_{aa}e_{ac}^2 + 24\sqrt{3}d_{aa}e_{ac}e_{bc} - 48\sqrt{3}d_{aa}e_{bc}^2 + 18d_{ab}e_{ab}^2 + 18d_{ab}e_{ab}e_{ac} - 18d_{ab}e_{ab}e_{bc} - 9d_{ab}e_{ac}^2 - 36d_{ab}e_{ac}e_{bc} - 9d_{ab}e_{bc}^2 - 9d_{ac}e_{ab}^2 + 18d_{ac}e_{ab}e_{ac} + 36d_{ac}e_{ab}e_{bc} + 18d_{ac}e_{ac}^2 + 18d_{ac}e_{ac}e_{bc} - 9d_{ac}e_{bc}^2 + 16d_{bb}^3 - 24d_{bb}^2 d_{cc} - 24d_{bb}d_{cc}^2 + 24\sqrt{3}d_{bb}e_{ab}^2 + 24\sqrt{3}d_{bb}e_{ab}e_{ac} + 48\sqrt{3}d_{bb}e_{ab}e_{bc} - 48\sqrt{3}d_{bb}e_{ac}^2 + 24\sqrt{3}d_{bb}e_{ac}e_{bc} + 24\sqrt{3}d_{bb}e_{bc}^2 - 9d_{bc}e_{ab}^2 - 36d_{bc}e_{ab}e_{ac} - 18d_{bc}e_{ab}e_{bc} - 9d_{bc}e_{ac}^2 + 18d_{bc}e_{ac}e_{bc} + 18d_{bc}e_{bc}^2 + 16d_{cc}^3 - 48\sqrt{3}d_{cc}e_{ab}^2 + 24\sqrt{3}d_{cc}e_{ab}e_{ac} - 24\sqrt{3}d_{cc}e_{ab}e_{bc} + 24\sqrt{3}d_{cc}e_{ac}^2 - 48\sqrt{3}d_{cc}e_{ac}e_{bc} + 24\sqrt{3}d_{cc}e_{bc}^2$ |
| $ReE_{3,9}$ | $27e_{ab}^2 e_{ac} + 27e_{ab}^2 e_{bc} + 27e_{ab}e_{ac}^2 - 27e_{ab}e_{bc}^2 - 27e_{ac}^2 e_{bc} - 27e_{ac}e_{bc}^2$ |
| $ReE_{4,1}$ | $18\sqrt{3}d_{ab}^2 d_{ac}e_{ab} - 18\sqrt{3}d_{ab}^2 d_{ac}e_{ac} + 18\sqrt{3}d_{ab}^2 d_{ac}e_{bc} - 18\sqrt{3}d_{ab}^2 d_{bc}e_{ab} + 18\sqrt{3}d_{ab}^2 d_{bc}e_{ac} - 18\sqrt{3}d_{ab}^2 d_{bc}e_{bc} - 18\sqrt{3}d_{ab}d_{ac}^2 e_{ab} + 18\sqrt{3}d_{ab}d_{ac}^2 e_{ac} - 18\sqrt{3}d_{ab}d_{ac}^2 e_{bc} + 18\sqrt{3}d_{ab}d_{bc}^2 e_{ab} - 18\sqrt{3}d_{ab}d_{bc}^2 e_{ac} + 18\sqrt{3}d_{ab}d_{bc}^2 e_{bc} + 18\sqrt{3}d_{ac}^2 d_{bc}e_{ab} - 18\sqrt{3}d_{ac}^2 d_{bc}e_{ac} + 18\sqrt{3}d_{ac}^2 d_{bc}e_{bc} - 18\sqrt{3}d_{ac}d_{bc}^2 e_{ab} + 18\sqrt{3}d_{ac}d_{bc}^2 e_{ac} - 18\sqrt{3}d_{ac}d_{bc}^2 e_{bc}$ |

$ReE_{4,4}$ $36d_{aa}^2d_{ab}^2 - 36d_{aa}^2d_{ab}d_{ac} - 36d_{aa}^2d_{ab}d_{bc} + 36d_{aa}^2d_{ac}^2 - 36d_{aa}^2d_{ac}d_{bc} + 36d_{aa}^2d_{bc}^2 - 36d_{aa}d_{ab}^2d_{bb} - 36d_{aa}d_{ab}^2d_{cc} + 36d_{aa}d_{ab}d_{ac}d_{bb} + 36d_{aa}d_{ab}d_{ac}d_{cc} + 36d_{aa}d_{ab}d_{bb}d_{bc} + 36d_{aa}d_{ab}d_{bc}d_{cc} - 36d_{aa}d_{ac}^2d_{bb} - 36d_{aa}d_{ac}^2d_{cc} + 36d_{aa}d_{ac}d_{bb}d_{bc} + 36d_{aa}d_{ac}d_{bc}d_{cc} - 36d_{aa}d_{bb}d_{bc}^2 - 36d_{aa}d_{bc}^2d_{cc} + 36d_{ab}^2d_{bb}^2 - 36d_{ab}^2d_{bb}d_{cc} + 36d_{ab}^2d_{cc}^2 + 36\sqrt{3}d_{ab}^2e_{ab}^2 - 18\sqrt{3}d_{ab}^2e_{ab}e_{ac} + 18\sqrt{3}d_{ab}^2e_{ab}e_{bc} - 18\sqrt{3}d_{ab}^2e_{ac}^2 + 36\sqrt{3}d_{ab}^2e_{ac}e_{bc} - 18\sqrt{3}d_{ab}^2e_{bc}^2 - 36d_{ab}d_{ac}d_{bb}^2 + 36d_{ab}d_{ac}d_{bb}d_{cc} - 36d_{ab}d_{ac}d_{cc}^2 - 36\sqrt{3}d_{ab}d_{ac}e_{ab}^2 + 72\sqrt{3}d_{ab}d_{ac}e_{ab}e_{ac} + 36\sqrt{3}d_{ab}d_{ac}e_{ab}e_{bc} - 36\sqrt{3}d_{ab}d_{ac}e_{ac}^2 - 36\sqrt{3}d_{ab}d_{ac}e_{ac}e_{bc} + 72\sqrt{3}d_{ab}d_{ac}e_{bc}^2 - 36d_{ab}d_{bb}^2d_{bc} + 36d_{ab}d_{bb}d_{bc}d_{cc} - 36d_{ab}d_{bc}d_{cc}^2 - 36\sqrt{3}d_{ab}d_{bc}e_{ab}^2 - 36\sqrt{3}d_{ab}d_{bc}e_{ab}e_{ac} - 72\sqrt{3}d_{ab}d_{bc}e_{ab}e_{bc} + 72\sqrt{3}d_{ab}d_{bc}e_{ac}^2 - 36\sqrt{3}d_{ab}d_{bc}e_{ac}e_{bc} - 36\sqrt{3}d_{ab}d_{bc}e_{bc}^2 + 36d_{ac}^2d_{bb}^2 - 36d_{ac}^2d_{bb}d_{cc} + 36d_{ac}^2d_{cc}^2 - 18\sqrt{3}d_{ac}^2e_{ab}^2 - 18\sqrt{3}d_{ac}^2e_{ab}e_{ac} - 36\sqrt{3}d_{ac}^2e_{ab}e_{bc} + 36\sqrt{3}d_{ac}^2e_{ac}^2 - 18\sqrt{3}d_{ac}^2e_{ac}e_{bc} - 18\sqrt{3}d_{ac}^2e_{bc}^2 - 36d_{ac}d_{bb}^2d_{bc} + 36d_{ac}d_{bb}d_{bc}d_{cc} - 36d_{ac}d_{bc}^2d_{cc} + 72\sqrt{3}d_{ac}d_{bc}e_{ab}^2 - 36\sqrt{3}d_{ac}d_{bc}e_{ab}e_{ac} + 36\sqrt{3}d_{ac}d_{bc}e_{ab}e_{bc} - 36\sqrt{3}d_{ac}d_{bc}e_{ac}^2 + 72\sqrt{3}d_{ac}d_{bc}e_{ac}e_{bc} - 36\sqrt{3}d_{ac}d_{bc}e_{bc}^2 + 36d_{bb}^2d_{bc}^2 - 36d_{bb}d_{bc}^2d_{cc} + 36d_{bc}^2d_{cc}^2 - 18\sqrt{3}d_{bc}^2e_{ab}^2 + 36\sqrt{3}d_{bc}^2e_{ab}e_{ac} + 18\sqrt{3}d_{bc}^2e_{ab}e_{bc} - 18\sqrt{3}d_{bc}^2e_{ac}^2 - 18\sqrt{3}d_{bc}^2e_{ac}e_{bc} + 36\sqrt{3}d_{bc}^2e_{bc}^2 + 48e_{ab}^4 - 192e_{ab}^3e_{ac} + 192e_{ab}^3e_{bc} + 288e_{ab}^2e_{ac}^2 - 576e_{ab}^2e_{ac}e_{bc} + 288e_{ab}^2e_{bc}^2 - 192e_{ab}e_{ac}^3 + 576e_{ab}e_{ac}^2e_{bc} - 576e_{ab}e_{ac}e_{bc}^2 + 192e_{ab}e_{bc}^3 + 48e_{ac}^4 - 192e_{ac}^3e_{bc} + 288e_{ac}^2e_{bc}^2 - 192e_{ac}e_{bc}^3 + 48e_{bc}^4$

$ReE_{4,7}$ $108d_{aa}^2d_{ab}e_{ac} + 108d_{aa}^2d_{ab}e_{bc} + 108d_{aa}^2d_{ac}e_{ab} - 108d_{aa}^2d_{ac}e_{bc} - 144\sqrt{3}d_{aa}^2d_{bb}e_{ab} + 144\sqrt{3}d_{aa}^2d_{bb}e_{ac} - 144\sqrt{3}d_{aa}^2d_{bb}e_{bc} - 108d_{aa}^2d_{bc}e_{ab} - 108d_{aa}^2d_{bc}e_{ac} + 144\sqrt{3}d_{aa}^2d_{cc}e_{ab} - 144\sqrt{3}d_{aa}^2d_{cc}e_{ac} + 144\sqrt{3}d_{aa}^2d_{cc}e_{bc} - 108d_{aa}d_{ab}d_{bb}e_{ac} - 108d_{aa}d_{ab}d_{bb}e_{bc} - 108d_{aa}d_{ab}d_{cc}e_{ac} - 108d_{aa}d_{ab}d_{cc}e_{bc} - 108d_{aa}d_{ac}d_{bb}e_{ab} + 108d_{aa}d_{ac}d_{bb}e_{bc} - 108d_{aa}d_{ac}d_{cc}e_{ab} + 108d_{aa}d_{ac}d_{cc}e_{bc} + 144\sqrt{3}d_{aa}d_{bb}^2e_{ab} - 144\sqrt{3}d_{aa}d_{bb}^2e_{ac} + 144\sqrt{3}d_{aa}d_{bb}^2e_{bc} + 108d_{aa}d_{bb}d_{bc}e_{ab} + 108d_{aa}d_{bb}d_{bc}e_{ac} + 108d_{aa}d_{bc}d_{cc}e_{ab} + 108d_{aa}d_{bc}d_{cc}e_{ac} - 144\sqrt{3}d_{aa}d_{cc}^2e_{ab} + 144\sqrt{3}d_{aa}d_{cc}^2e_{ac} - 144\sqrt{3}d_{aa}d_{cc}^2e_{bc} + 108d_{ab}d_{bb}^2e_{ac} + 108d_{ab}d_{bb}^2e_{bc} - 108d_{ab}d_{bb}d_{cc}e_{ac} - 108d_{ab}d_{bb}d_{cc}e_{bc} + 108d_{ab}d_{cc}^2e_{ac} + 108d_{ab}d_{cc}^2e_{bc} + 108\sqrt{3}d_{ab}e_{ab}^2e_{ac} + 108\sqrt{3}d_{ab}e_{ab}^2e_{bc} - 54\sqrt{3}d_{ab}e_{ab}e_{ac}^2 + 54\sqrt{3}d_{ab}e_{ab}e_{bc}^2 - 54\sqrt{3}d_{ab}e_{ac}^3 + 54\sqrt{3}d_{ab}e_{ac}^2e_{bc} + 54\sqrt{3}d_{ab}e_{ac}e_{bc}^2 - 54\sqrt{3}d_{ab}e_{bc}^3 + 108d_{ac}d_{bb}^2e_{ab} - 108d_{ac}d_{bb}^2e_{bc} - 108d_{ac}d_{bb}d_{cc}e_{ab} + 108d_{ac}d_{bb}d_{cc}e_{bc} + 108d_{ac}d_{cc}^2e_{ab} - 108d_{ac}d_{cc}^2e_{bc} - 54\sqrt{3}d_{ac}e_{ab}^3 - 54\sqrt{3}d_{ac}e_{ab}^2e_{ac} - 54\sqrt{3}d_{ac}e_{ab}^2e_{bc} + 108\sqrt{3}d_{ac}e_{ab}e_{ac}^2 + 54\sqrt{3}d_{ac}e_{ab}e_{bc}^2 - 108\sqrt{3}d_{ac}e_{ac}^2e_{bc} + 54\sqrt{3}d_{ac}e_{ac}e_{bc}^2 + 54\sqrt{3}d_{ac}e_{bc}^3 - 108d_{bb}^2d_{bc}e_{ab} - 108d_{bb}^2d_{bc}e_{ac} - 144\sqrt{3}d_{bb}^2d_{cc}e_{ab} + 144\sqrt{3}d_{bb}^2d_{cc}e_{ac} - 144\sqrt{3}d_{bb}^2d_{cc}e_{bc} + 108d_{bb}d_{bc}d_{cc}e_{ab} + 108d_{bb}d_{bc}d_{cc}e_{ac} + 144\sqrt{3}d_{bb}d_{cc}^2e_{ab} - 144\sqrt{3}d_{bb}d_{cc}^2e_{ac} + 144\sqrt{3}d_{bb}d_{cc}^2e_{bc} - 108d_{bc}d_{cc}^2e_{ab} - 108d_{bc}d_{cc}^2e_{ac} + 54\sqrt{3}d_{bc}e_{ab}^3 - 54\sqrt{3}d_{bc}e_{ab}^2e_{ac} - 54\sqrt{3}d_{bc}e_{ab}^2e_{bc} - 54\sqrt{3}d_{bc}e_{ab}e_{ac}^2 - 108\sqrt{3}d_{bc}e_{ab}e_{bc}^2 + 54\sqrt{3}d_{bc}e_{ac}^3 + 54\sqrt{3}d_{bc}e_{ac}^2e_{bc} - 108\sqrt{3}d_{bc}e_{ac}e_{bc}^2$

$ReE_{4,10}$ $108d_{aa}^2e_{ab}^2 + 108d_{aa}^2e_{ab}e_{ac} - 108d_{aa}^2e_{ab}e_{bc} + 108d_{aa}^2e_{ac}^2 + 108d_{aa}^2e_{ac}e_{bc} + 108d_{aa}^2e_{bc}^2 - 108d_{aa}d_{bb}e_{ab}^2 - 108d_{aa}d_{bb}e_{ab}e_{ac} + 108d_{aa}d_{bb}e_{ab}e_{bc} - 108d_{aa}d_{bb}e_{ac}^2 - 108d_{aa}d_{bb}e_{ac}e_{bc} - 108d_{aa}d_{bb}e_{bc}^2 - 108d_{aa}d_{cc}e_{ab}^2 - 108d_{aa}d_{cc}e_{ab}e_{ac} + 108d_{aa}d_{cc}e_{ab}e_{bc} - 108d_{aa}d_{cc}e_{ac}^2 - 108d_{aa}d_{cc}e_{ac}e_{bc} - 108d_{aa}d_{cc}e_{bc}^2 + 108d_{bb}^2e_{ab}^2 + 108d_{bb}^2e_{ab}e_{ac} - 108d_{bb}^2e_{ab}e_{bc} + 108d_{bb}^2e_{ac}^2 + 108d_{bb}^2e_{ac}e_{bc} + 108d_{bb}^2e_{bc}^2 - 108d_{bb}d_{cc}e_{ab}^2 - 108d_{bb}d_{cc}e_{ab}e_{ac} + 108d_{bb}d_{cc}e_{ab}e_{bc} - 108d_{bb}d_{cc}e_{ac}^2 - 108d_{bb}d_{cc}e_{ac}e_{bc} - 108d_{bb}d_{cc}e_{bc}^2 + 108d_{cc}^2e_{ab}^2 + 108d_{cc}^2e_{ab}e_{ac} - 108d_{cc}^2e_{ab}e_{bc} + 108d_{cc}^2e_{ac}^2 + 108d_{cc}^2e_{ac}e_{bc} + 108d_{cc}^2e_{bc}^2 - 36\sqrt{3}e_{ab}^4 - 18\sqrt{3}e_{ab}^3e_{ac} + 18\sqrt{3}e_{ab}^3e_{bc} + 108\sqrt{3}e_{ab}^2e_{ac}^2 + 108\sqrt{3}e_{ab}^2e_{ac}e_{bc} + 108\sqrt{3}e_{ab}^2e_{bc}^2 - 18\sqrt{3}e_{ab}e_{ac}^3 - 108\sqrt{3}e_{ab}e_{ac}^2e_{bc} + 108\sqrt{3}e_{ab}e_{ac}e_{bc}^2 + 18\sqrt{3}e_{ab}e_{bc}^3 - 36\sqrt{3}e_{ac}^4 - 18\sqrt{3}e_{ac}^3e_{bc} + 108\sqrt{3}e_{ac}^2e_{bc}^2 - 18\sqrt{3}e_{ac}e_{bc}^3 - 36\sqrt{3}e_{bc}^4$

$ReE_{5,2}$ (105 terms; see the full text file)

$ReE_{5,5}$ (180 terms; see the full text file)

$ReE_{5,8}$ (252 terms; see the full text file)

$ReE_{5,11}$ (120 terms; see the full text file)

$ReE_{5,14}$ $-54d_{aa}e_{ab}^4 - 270d_{aa}e_{ab}^3e_{ac} - 54d_{aa}e_{ab}^3e_{bc} - 324d_{aa}e_{ab}^2e_{ac}^2 + 162d_{aa}e_{ab}^2e_{ac}e_{bc} + 162d_{aa}e_{ab}^2e_{bc}^2 - 270d_{aa}e_{ab}e_{ac}^3 - 162d_{aa}e_{ab}e_{ac}^2e_{bc} - 324d_{aa}e_{ab}e_{ac}e_{bc}^2 - 216d_{aa}e_{ab}e_{bc}^3 - 54d_{aa}e_{ac}^4 + 54d_{aa}e_{ac}^3e_{bc} + 162d_{aa}e_{ac}^2e_{bc}^2 + 216d_{aa}e_{ac}e_{bc}^3 + 108d_{aa}e_{bc}^4 - 54d_{bb}e_{ab}^4 + 54d_{bb}e_{ab}^3e_{ac} + 270d_{bb}e_{ab}^3e_{bc} + 162d_{bb}e_{ab}^2e_{ac}^2 + 162d_{bb}e_{ab}^2e_{ac}e_{bc} - 324d_{bb}e_{ab}^2e_{bc}^2 + 216d_{bb}e_{ab}e_{ac}^3 + 324d_{bb}e_{ab}e_{ac}^2e_{bc} + 162d_{bb}e_{ab}e_{ac}e_{bc}^2 + 270d_{bb}e_{ab}e_{bc}^3 + 108d_{bb}e_{ac}^4 + 216d_{bb}e_{ac}^3e_{bc} + 162d_{bb}e_{ac}^2e_{bc}^2 + 54d_{bb}e_{ac}e_{bc}^3 - 54d_{bb}e_{bc}^4 + 108d_{cc}e_{ab}^4 + 216d_{cc}e_{ab}^3e_{ac} - 216d_{cc}e_{ab}^3e_{bc} + 162d_{cc}e_{ab}^2e_{ac}^2 - 324d_{cc}e_{ab}^2e_{ac}e_{bc} + 162d_{cc}e_{ab}^2e_{bc}^2 + 54d_{cc}e_{ab}e_{ac}^3 - 162d_{cc}e_{ab}e_{ac}^2e_{bc} + 162d_{cc}e_{ab}e_{ac}e_{bc}^2 - 54d_{cc}e_{ab}e_{bc}^3 - 54d_{cc}e_{ac}^4 - 270d_{cc}e_{ac}^3e_{bc} - 324d_{cc}e_{ac}^2e_{bc}^2 - 270d_{cc}e_{ac}e_{bc}^3 - 54d_{cc}e_{bc}^4$

$ReE_{6,0}$ $d_{ab}^6 - 3d_{ab}^5d_{ac} - 3d_{ab}^5d_{bc} + 6d_{ab}^4d_{ac}^2 + 3d_{ab}^4d_{ac}d_{bc} + 6d_{ab}^4d_{bc}^2 - 7d_{ab}^3d_{ac}^3 - 3d_{ab}^3d_{ac}^2d_{bc} - 3d_{ab}^3d_{ac}d_{bc}^2 - 7d_{ab}^3d_{bc}^3 + 6d_{ab}^2d_{ac}^4 - 3d_{ab}^2d_{ac}^3d_{bc} + 9d_{ab}^2d_{ac}^2d_{bc}^2 - 3d_{ab}^2d_{ac}d_{bc}^3 + 6d_{ab}^2d_{bc}^4 - 3d_{ab}d_{ac}^5 + 3d_{ab}d_{ac}^4d_{bc} - 3d_{ab}d_{ac}^3d_{bc}^2 - 3d_{ab}d_{ac}^2d_{bc}^3 + 3d_{ab}d_{ac}d_{bc}^4 - 3d_{ab}d_{bc}^5 + d_{ac}^6 - 3d_{ac}^5d_{bc} + 6d_{ac}^4d_{bc}^2 - 7d_{ac}^3d_{bc}^3 + 6d_{ac}^2d_{bc}^4 - 3d_{ac}d_{bc}^5 + d_{bc}^6$

$ReE_{6,3}$ (222 terms; see the full text file)

$ReE_{6,6}$ (839 terms; see the full text file)

$ReE_{6,9}$ (786 terms; see the full text file)

$ReE_{6,12}$ (623 terms; see the full text file)

$ReE_{6,15}$ (162 terms; see the full text file)

$ReE_{6,18}$ $27e_{ab}^6 + 81e_{ab}^5e_{ac} - 81e_{ab}^5e_{bc} + 162e_{ab}^4e_{ac}^2 - 81e_{ab}^4e_{ac}e_{bc} + 162e_{ab}^4e_{bc}^2 + 189e_{ab}^3e_{ac}^3 - 81e_{ab}^3e_{ac}^2e_{bc} + 81e_{ab}^3e_{ac}e_{bc}^2 - 189e_{ab}^3e_{bc}^3 + 162e_{ab}^2e_{ac}^4 + 81e_{ab}^2e_{ac}^3e_{bc} + 243e_{ab}^2e_{ac}^2e_{bc}^2 + 81e_{ab}^2e_{ac}e_{bc}^3 + 162e_{ab}^2e_{bc}^4 + 81e_{ab}e_{ac}^5 + 81e_{ab}e_{ac}^4e_{bc} + 81e_{ab}e_{ac}^3e_{bc}^2 - 81e_{ab}e_{ac}^2e_{bc}^3 - 81e_{ab}e_{ac}e_{bc}^4 - 81e_{ab}e_{bc}^5 + 27e_{ac}^6 + 81e_{ac}^5e_{bc} + 162e_{ac}^4e_{bc}^2 + 189e_{ac}^3e_{bc}^3 + 162e_{ac}^2e_{bc}^4 + 81e_{ac}e_{bc}^5 + 27e_{bc}^6$

## 3 $S(2) \times S(1)$ acting on $(a, b, p)$, $k = 2$

$|G| = 2$, $m = 9$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 11 (plain Bursarial: 26; quotient dimension 5).
Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{pp}$
- inv[2] $= 2d_{ab}$
- inv[3] $= d_{ap} + d_{bp}$
- inv[4] $= e_{ap} + e_{bp}$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 1 | $d_{aa} - d_{bb}$ | $2e_{ab}$ |
| 1 | 1 | $d_{ap} - d_{bp}$ | $e_{ap} - e_{bp}$ |

| output | expanded |
|---|---|
| $inv[0]$ | $d_{aa} + d_{bb}$ |
| $inv[1]$ | $2d_{pp}$ |
| $inv[2]$ | $2d_{ab}$ |
| $inv[3]$ | $d_{ap} + d_{bp}$ |
| $inv[4]$ | $e_{ap} + e_{bp}$ |
| $ReE_{2,0}$ | $-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + 4e_{ab}^2$ |
| $ImE_{2,0}$ | $-4d_{aa}e_{ab} + 4d_{bb}e_{ab}$ |
| $ReE_{2,1}$ | $-2d_{aa}d_{ap} + 2d_{aa}d_{bp} + 2d_{ap}d_{bb} - 2d_{bb}d_{bp} + 4e_{ab}e_{ap} - 4e_{ab}e_{bp}$ |
| $ImE_{2,1}$ | $-2d_{aa}e_{ap} + 2d_{aa}e_{bp} - 4d_{ap}e_{ab} + 2d_{bb}e_{ap} - 2d_{bb}e_{bp} + 4d_{bp}e_{ab}$ |
| $ReE_{2,2}$ | $-d_{ap}^2 + 2d_{ap}d_{bp} - d_{bp}^2 + e_{ap}^2 - 2e_{ap}e_{bp} + e_{bp}^2$ |
| $ImE_{2,2}$ | $-2d_{ap}e_{ap} + 2d_{ap}e_{bp} + 2d_{bp}e_{ap} - 2d_{bp}e_{bp}$ |

# 4 $S(2)$ acting on $(a, b)$, $k = 1$

$|G| = 2$, $m = 5$ slots; $h = (ab)$ (order $r = 2$), $s =$ none, mode scalar; complex coefficients. Outputs: 5 (plain Bursarial: 14; quotient dimension 2).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{ab}$
- inv[2] $= e_a + e_b$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 1 | $d_{aa} - d_{bb}$ | $e_a - e_b$ |

| output | expanded |
|---|---|
| $inv[0]$ | $d_{aa} + d_{bb}$ |
| $inv[1]$ | $2d_{ab}$ |
| $inv[2]$ | $e_a + e_b$ |
| $ReE_{2,0}$ | $-d_{aa}^2 + 2d_{aa}d_{bb} - d_{bb}^2 + e_a^2 - 2e_ae_b + e_b^2$ |
| $ImE_{2,0}$ | $-2d_{aa}e_a + 2d_{aa}e_b + 2d_{bb}e_a - 2d_{bb}e_b$ |

# 5 $S(3)$ acting on $(a, b, c)$, $k = 1$

$|G| = 6$, $m = 9$ slots; $h = (abc)$ (order $r = 3$), $s = (bc)$, mode x; real coefficients. Outputs: 24 (plain Bursarial: 174; quotient dimension 3).

Invariants output directly:

- inv[0] $= 2d_{aa} + 2d_{bb} + 2d_{cc}$
- inv[1] $= 2d_{ab} + 2d_{ac} + 2d_{bc}$
- inv[2] $= 2e_a + 2e_b + 2e_c$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 1 | $2d_{aa} - d_{bb} - d_{cc}$ | $\sqrt{3}\,(-d_{bb} + d_{cc})$ |
| 1 | 2 | $2e_a - e_b - e_c$ | $\sqrt{3}\,(e_b - e_c)$ |
| 2 | | 0 | 0 |
| 3 | 1 | $\frac{d_{ab}}{2} + \frac{d_{ac}}{2} - d_{bc}$ | $\frac{\sqrt{3}(-d_{ab}+d_{ac})}{2}$ |

| output | expanded |
|---|---|
| $inv[0]$ | $2d_{aa} + 2d_{bb} + 2d_{cc}$ |
| $inv[1]$ | $2d_{ab} + 2d_{ac} + 2d_{bc}$ |
| $inv[2]$ | $2e_a + 2e_b + 2e_c$ |
| $ReE_{2,1}$ | $-24d_{aa}e_a + 12d_{aa}e_b + 12d_{aa}e_c + 12d_{bb}e_a - 24d_{bb}e_b + 12d_{bb}e_c + 12d_{cc}e_a + 12d_{cc}e_b - 24d_{cc}e_c$ |
| $ReE_{2,4}$ | $-6d_{ab}e_a - 6d_{ab}e_b + 12d_{ab}e_c - 6d_{ac}e_a + 12d_{ac}e_b - 6d_{ac}e_c + 12d_{bc}e_a - 6d_{bc}e_b - 6d_{bc}e_c$ |
| $ReE_{3,0}$ | $16d_{aa}^3 - 24d_{aa}^2d_{bb} - 24d_{aa}^2d_{cc} - 24d_{aa}d_{bb}^2 + 96d_{aa}d_{bb}d_{cc} - 24d_{aa}d_{cc}^2 + 16d_{bb}^3 - 24d_{bb}^2d_{cc} - 24d_{bb}d_{cc}^2 + 16d_{cc}^3$ |
| $ReE_{3,3}$ | $12d_{aa}^2d_{ab} + 12d_{aa}^2d_{ac} - 24d_{aa}^2d_{bc} - 48d_{aa}d_{ab}d_{bb} + 24d_{aa}d_{ab}d_{cc} + 24d_{aa}d_{ac}d_{bb} - 48d_{aa}d_{ac}d_{cc} + 24d_{aa}d_{bb}d_{bc} + 24d_{aa}d_{bc}d_{cc} + 12d_{ab}d_{bb}^2 + 24d_{ab}d_{bb}d_{cc} - 24d_{ab}d_{cc}^2 - 24d_{ac}d_{bb}^2 + 24d_{ac}d_{bb}d_{cc} + 12d_{ac}d_{cc}^2 + 12d_{bb}^2d_{bc} - 48d_{bb}d_{bc}d_{cc} + 12d_{bc}d_{cc}^2 + 16e_a^3 - 24e_a^2e_b - 24e_a^2e_c - 24e_ae_b^2 + 96e_ae_be_c - 24e_ae_c^2 + 16e_b^3 - 24e_b^2e_c - 24e_be_c^2 + 16e_c^3$ |
| $ReE_{3,6}$ | $-6d_{aa}d_{ab}^2 + 24d_{aa}d_{ab}d_{ac} - 12d_{aa}d_{ab}d_{bc} - 6d_{aa}d_{ac}^2 - 12d_{aa}d_{ac}d_{bc} + 12d_{aa}d_{bc}^2 - 6d_{ab}^2d_{bb} + 12d_{ab}^2d_{cc} - 12d_{ab}d_{ac}d_{bb} - 12d_{ab}d_{ac}d_{cc} + 24d_{ab}d_{bb}d_{bc} - 12d_{ab}d_{bc}d_{cc} + 12d_{ac}^2d_{bb} - 6d_{ac}^2d_{cc} - 12d_{ac}d_{bb}d_{bc} + 24d_{ac}d_{bc}d_{cc} - 6d_{bb}d_{bc}^2 - 6d_{bc}^2d_{cc}$ |
| $ReE_{3,9}$ | $-2d_{ab}^3 + 3d_{ab}^2d_{ac} + 3d_{ab}^2d_{bc} + 3d_{ab}d_{ac}^2 - 12d_{ab}d_{ac}d_{bc} + 3d_{ab}d_{bc}^2 - 2d_{ac}^3 + 3d_{ac}^2d_{bc} + 3d_{ac}d_{bc}^2 - 2d_{bc}^3$ |
| $ReE_{4,2}$ | $144d_{aa}^2e_a^2 - 144d_{aa}^2e_ae_b - 144d_{aa}^2e_ae_c + 144d_{aa}^2e_b^2 - 144d_{aa}^2e_be_c + 144d_{aa}^2e_c^2 - 144d_{aa}d_{bb}e_a^2 + 144d_{aa}d_{bb}e_ae_b + 144d_{aa}d_{bb}e_ae_c - 144d_{aa}d_{bb}e_b^2 + 144d_{aa}d_{bb}e_be_c - 144d_{aa}d_{bb}e_c^2 - 144d_{aa}d_{cc}e_a^2 + 144d_{aa}d_{cc}e_ae_b + 144d_{aa}d_{cc}e_ae_c - 144d_{aa}d_{cc}e_b^2 + 144d_{aa}d_{cc}e_be_c - 144d_{aa}d_{cc}e_c^2 + 144d_{bb}^2e_a^2 - 144d_{bb}^2e_ae_b - 144d_{bb}^2e_ae_c + 144d_{bb}^2e_b^2 - 144d_{bb}^2e_be_c + 144d_{bb}^2e_c^2 - 144d_{bb}d_{cc}e_a^2 + 144d_{bb}d_{cc}e_ae_b + 144d_{bb}d_{cc}e_ae_c - 144d_{bb}d_{cc}e_b^2 + 144d_{bb}d_{cc}e_be_c - 144d_{bb}d_{cc}e_c^2 + 144d_{cc}^2e_a^2 - 144d_{cc}^2e_ae_b - 144d_{cc}^2e_ae_c + 144d_{cc}^2e_b^2 - 144d_{cc}^2e_be_c + 144d_{cc}^2e_c^2$ |

$ReE_{4,5}$ $72d_{aa}d_{ab}e_a^2 - 72d_{aa}d_{ab}e_ae_b - 72d_{aa}d_{ab}e_ae_c + 72d_{aa}d_{ab}e_b^2 - 72d_{aa}d_{ab}e_be_c + 72d_{aa}d_{ab}e_c^2 + 72d_{aa}d_{ac}e_a^2 - 72d_{aa}d_{ac}e_ae_b - 72d_{aa}d_{ac}e_ae_c + 72d_{aa}d_{ac}e_b^2 - 72d_{aa}d_{ac}e_be_c + 72d_{aa}d_{ac}e_c^2 - 144d_{aa}d_{bc}e_a^2 + 144d_{aa}d_{bc}e_ae_b + 144d_{aa}d_{bc}e_ae_c - 144d_{aa}d_{bc}e_b^2 + 144d_{aa}d_{bc}e_be_c - 144d_{aa}d_{bc}e_c^2 + 72d_{ab}d_{bb}e_a^2 - 72d_{ab}d_{bb}e_ae_b - 72d_{ab}d_{bb}e_ae_c + 72d_{ab}d_{bb}e_b^2 - 72d_{ab}d_{bb}e_be_c + 72d_{ab}d_{bb}e_c^2 - 144d_{ab}d_{cc}e_a^2 + 144d_{ab}d_{cc}e_ae_b + 144d_{ab}d_{cc}e_ae_c - 144d_{ab}d_{cc}e_b^2 + 144d_{ab}d_{cc}e_be_c - 144d_{ab}d_{cc}e_c^2 - 144d_{ac}d_{bb}e_a^2 + 144d_{ac}d_{bb}e_ae_b + 144d_{ac}d_{bb}e_ae_c - 144d_{ac}d_{bb}e_b^2 + 144d_{ac}d_{bb}e_be_c - 144d_{ac}d_{bb}e_c^2 + 72d_{ac}d_{cc}e_a^2 - 72d_{ac}d_{cc}e_ae_b - 72d_{ac}d_{cc}e_ae_c + 72d_{ac}d_{cc}e_b^2 - 72d_{ac}d_{cc}e_be_c + 72d_{ac}d_{cc}e_c^2 + 72d_{bb}d_{bc}e_a^2 - 72d_{bb}d_{bc}e_ae_b - 72d_{bb}d_{bc}e_ae_c + 72d_{bb}d_{bc}e_b^2 - 72d_{bb}d_{bc}e_be_c + 72d_{bb}d_{bc}e_c^2 + 72d_{bc}d_{cc}e_a^2 - 72d_{bc}d_{cc}e_ae_b - 72d_{bc}d_{cc}e_ae_c + 72d_{bc}d_{cc}e_b^2 - 72d_{bc}d_{cc}e_be_c + 72d_{bc}d_{cc}e_c^2$

$ReE_{4,8}$ $36d_{ab}^2e_a^2 - 36d_{ab}^2e_ae_b - 36d_{ab}^2e_ae_c + 36d_{ab}^2e_b^2 - 36d_{ab}^2e_be_c + 36d_{ab}^2e_c^2 - 36d_{ab}d_{ac}e_a^2 + 36d_{ab}d_{ac}e_ae_b + 36d_{ab}d_{ac}e_ae_c - 36d_{ab}d_{ac}e_b^2 + 36d_{ab}d_{ac}e_be_c - 36d_{ab}d_{ac}e_c^2 - 36d_{ab}d_{bc}e_a^2 + 36d_{ab}d_{bc}e_ae_b + 36d_{ab}d_{bc}e_ae_c - 36d_{ab}d_{bc}e_b^2 + 36d_{ab}d_{bc}e_be_c - 36d_{ab}d_{bc}e_c^2 + 36d_{ac}^2e_a^2 - 36d_{ac}^2e_ae_b - 36d_{ac}^2e_ae_c + 36d_{ac}^2e_b^2 - 36d_{ac}^2e_be_c + 36d_{ac}^2e_c^2 - 36d_{ac}d_{bc}e_a^2 + 36d_{ac}d_{bc}e_ae_b + 36d_{ac}d_{bc}e_ae_c - 36d_{ac}d_{bc}e_b^2 + 36d_{ac}d_{bc}e_be_c - 36d_{ac}d_{bc}e_c^2 + 36d_{bc}^2e_a^2 - 36d_{bc}^2e_ae_b - 36d_{bc}^2e_ae_c + 36d_{bc}^2e_b^2 - 36d_{bc}^2e_be_c + 36d_{bc}^2e_c^2$

$ReE_{5,1}$ $-192d_{aa}^4e_a + 96d_{aa}^4e_b + 96d_{aa}^4e_c + 384d_{aa}^3d_{bb}e_a + 96d_{aa}^3d_{bb}e_b - 480d_{aa}^3d_{bb}e_c + 384d_{aa}^3d_{cc}e_a - 480d_{aa}^3d_{cc}e_b + 96d_{aa}^3d_{cc}e_c - 288d_{aa}^2d_{bb}^2e_a - 288d_{aa}^2d_{bb}^2e_b + 576d_{aa}^2d_{bb}^2e_c - 576d_{aa}^2d_{bb}d_{cc}e_a + 288d_{aa}^2d_{bb}d_{cc}e_b + 288d_{aa}^2d_{bb}d_{cc}e_c - 288d_{aa}^2d_{cc}^2e_a + 576d_{aa}^2d_{cc}^2e_b - 288d_{aa}^2d_{cc}^2e_c + 96d_{aa}d_{bb}^3e_a + 384d_{aa}d_{bb}^3e_b - 480d_{aa}d_{bb}^3e_c + 288d_{aa}d_{bb}^2d_{cc}e_a - 576d_{aa}d_{bb}^2d_{cc}e_b + 288d_{aa}d_{bb}^2d_{cc}e_c + 288d_{aa}d_{bb}d_{cc}^2e_a + 288d_{aa}d_{bb}d_{cc}^2e_b - 576d_{aa}d_{bb}d_{cc}^2e_c + 96d_{aa}d_{cc}^3e_a - 480d_{aa}d_{cc}^3e_b + 384d_{aa}d_{cc}^3e_c + 96d_{bb}^4e_a - 192d_{bb}^4e_b + 96d_{bb}^4e_c - 480d_{bb}^3d_{cc}e_a + 384d_{bb}^3d_{cc}e_b + 96d_{bb}^3d_{cc}e_c + 576d_{bb}^2d_{cc}^2e_a - 288d_{bb}^2d_{cc}^2e_b - 288d_{bb}^2d_{cc}^2e_c - 480d_{bb}d_{cc}^3e_a + 96d_{bb}d_{cc}^3e_b + 384d_{bb}d_{cc}^3e_c + 96d_{cc}^4e_a + 96d_{cc}^4e_b - 192d_{cc}^4e_c$

$ReE_{5,4}$ (135 terms; see the full text file)

$ReE_{5,7}$ (153 terms; see the full text file)

$ReE_{5,10}$ $-12d_{aa}d_{ab}^3e_a + 60d_{aa}d_{ab}^3e_b - 48d_{aa}d_{ab}^3e_c - 36d_{aa}d_{ab}^2d_{ac}e_a - 36d_{aa}d_{ab}^2d_{ac}e_b + 72d_{aa}d_{ab}^2d_{ac}e_c + 72d_{aa}d_{ab}^2d_{bc}e_a - 144d_{aa}d_{ab}^2d_{bc}e_b + 72d_{aa}d_{ab}^2d_{bc}e_c - 36d_{aa}d_{ab}d_{ac}^2e_a + 72d_{aa}d_{ab}d_{ac}^2e_b - 36d_{aa}d_{ab}d_{ac}^2e_c + 144d_{aa}d_{ab}d_{ac}d_{bc}e_a - 72d_{aa}d_{ab}d_{ac}d_{bc}e_b - 72d_{aa}d_{ab}d_{ac}d_{bc}e_c - 144d_{aa}d_{ab}d_{bc}^2e_a + 180d_{aa}d_{ab}d_{bc}^2e_b - 36d_{aa}d_{ab}d_{bc}^2e_c - 12d_{aa}d_{ac}^3e_a - 48d_{aa}d_{ac}^3e_b + 60d_{aa}d_{ac}^3e_c + 72d_{aa}d_{ac}^2d_{bc}e_a + 72d_{aa}d_{ac}^2d_{bc}e_b - 144d_{aa}d_{ac}^2d_{bc}e_c - 144d_{aa}d_{ac}d_{bc}^2e_a - 36d_{aa}d_{ac}d_{bc}^2e_b + 180d_{aa}d_{ac}d_{bc}^2e_c + 96d_{aa}d_{bc}^3e_a - 48d_{aa}d_{bc}^3e_b - 48d_{aa}d_{bc}^3e_c + 60d_{ab}^3d_{bb}e_a - 12d_{ab}^3d_{bb}e_b - 48d_{ab}^3d_{bb}e_c - 48d_{ab}^3d_{cc}e_a - 48d_{ab}^3d_{cc}e_b + 96d_{ab}^3d_{cc}e_c - 144d_{ab}^2d_{ac}d_{bb}e_a + 72d_{ab}^2d_{ac}d_{bb}e_b + 72d_{ab}^2d_{ac}d_{bb}e_c + 180d_{ab}^2d_{ac}d_{cc}e_a - 36d_{ab}^2d_{ac}d_{cc}e_b - 144d_{ab}^2d_{ac}d_{cc}e_c - 36d_{ab}^2d_{bb}d_{bc}e_a - 36d_{ab}^2d_{bb}d_{bc}e_b + 72d_{ab}^2d_{bb}d_{bc}e_c - 36d_{ab}^2d_{bc}d_{cc}e_a + 180d_{ab}^2d_{bc}d_{cc}e_b - 144d_{ab}^2d_{bc}d_{cc}e_c + 180d_{ab}d_{ac}^2d_{bb}e_a - 144d_{ab}d_{ac}^2d_{bb}e_b - 36d_{ab}d_{ac}^2d_{bb}e_c - 144d_{ab}d_{ac}^2d_{cc}e_a + 72d_{ab}d_{ac}^2d_{cc}e_b + 72d_{ab}d_{ac}^2d_{cc}e_c - 72d_{ab}d_{ac}d_{bb}d_{bc}e_a + 144d_{ab}d_{ac}d_{bb}d_{bc}e_b - 72d_{ab}d_{ac}d_{bb}d_{bc}e_c - 72d_{ab}d_{ac}d_{bc}d_{cc}e_a - 72d_{ab}d_{ac}d_{bc}d_{cc}e_b + 144d_{ab}d_{ac}d_{bc}d_{cc}e_c + 72d_{ab}d_{bb}d_{bc}^2e_a - 36d_{ab}d_{bb}d_{bc}^2e_b - 36d_{ab}d_{bb}d_{bc}^2e_c + 72d_{ab}d_{bc}^2d_{cc}e_a - 144d_{ab}d_{bc}^2d_{cc}e_b + 72d_{ab}d_{bc}^2d_{cc}e_c - 48d_{ac}^3d_{bb}e_a + 96d_{ac}^3d_{bb}e_b - 48d_{ac}^3d_{bb}e_c + 60d_{ac}^3d_{cc}e_a - 48d_{ac}^3d_{cc}e_b - 12d_{ac}^3d_{cc}e_c - 36d_{ac}^2d_{bb}d_{bc}e_a - 144d_{ac}^2d_{bb}d_{bc}e_b + 180d_{ac}^2d_{bb}d_{bc}e_c - 36d_{ac}^2d_{bc}d_{cc}e_a + 72d_{ac}^2d_{bc}d_{cc}e_b - 36d_{ac}^2d_{bc}d_{cc}e_c + 72d_{ac}d_{bb}d_{bc}^2e_a + 72d_{ac}d_{bb}d_{bc}^2e_b - 144d_{ac}d_{bb}d_{bc}^2e_c + 72d_{ac}d_{bc}^2d_{cc}e_a - 36d_{ac}d_{bc}^2d_{cc}e_b - 36d_{ac}d_{bc}^2d_{cc}e_c - 48d_{bb}d_{bc}^3e_a - 12d_{bb}d_{bc}^3e_b + 60d_{bb}d_{bc}^3e_c - 48d_{bc}^3d_{cc}e_a + 60d_{bc}^3d_{cc}e_b - 12d_{bc}^3d_{cc}e_c$

$ReE_{5,13}$ $6d_{ab}^4e_a + 6d_{ab}^4e_b - 12d_{ab}^4e_c - 30d_{ab}^3d_{ac}e_a + 6d_{ab}^3d_{ac}e_b + 24d_{ab}^3d_{ac}e_c + 6d_{ab}^3d_{bc}e_a - 30d_{ab}^3d_{bc}e_b + 24d_{ab}^3d_{bc}e_c + 36d_{ab}^2d_{ac}^2e_a - 18d_{ab}^2d_{ac}^2e_b - 18d_{ab}^2d_{ac}^2e_c + 18d_{ab}^2d_{ac}d_{bc}e_a + 18d_{ab}^2d_{ac}d_{bc}e_b - 36d_{ab}^2d_{ac}d_{bc}e_c - 18d_{ab}^2d_{bc}^2e_a + 36d_{ab}^2d_{bc}^2e_b - 18d_{ab}^2d_{bc}^2e_c - 30d_{ab}d_{ac}^3e_a + 24d_{ab}d_{ac}^3e_b + 6d_{ab}d_{ac}^3e_c + 18d_{ab}d_{ac}^2d_{bc}e_a - 36d_{ab}d_{ac}^2d_{bc}e_b + 18d_{ab}d_{ac}^2d_{bc}e_c - 36d_{ab}d_{ac}d_{bc}^2e_a + 18d_{ab}d_{ac}d_{bc}^2e_b + 18d_{ab}d_{ac}d_{bc}^2e_c + 24d_{ab}d_{bc}^3e_a - 30d_{ab}d_{bc}^3e_b + 6d_{ab}d_{bc}^3e_c + 6d_{ac}^4e_a - 12d_{ac}^4e_b + 6d_{ac}^4e_c + 6d_{ac}^3d_{bc}e_a + 24d_{ac}^3d_{bc}e_b - 30d_{ac}^3d_{bc}e_c - 18d_{ac}^2d_{bc}^2e_a - 18d_{ac}^2d_{bc}^2e_b + 36d_{ac}^2d_{bc}^2e_c + 24d_{ac}d_{bc}^3e_a + 6d_{ac}d_{bc}^3e_b - 30d_{ac}d_{bc}^3e_c - 12d_{bc}^4e_a + 6d_{bc}^4e_b + 6d_{bc}^4e_c$

$ReE_{6,0}$ $64d_{aa}^6 - 192d_{aa}^5d_{bb} - 192d_{aa}^5d_{cc} + 384d_{aa}^4d_{bb}^2 + 192d_{aa}^4d_{bb}d_{cc} + 384d_{aa}^4d_{cc}^2 - 448d_{aa}^3d_{bb}^3 - 192d_{aa}^3d_{bb}^2d_{cc} - 192d_{aa}^3d_{bb}d_{cc}^2 - 448d_{aa}^3d_{cc}^3 + 384d_{aa}^2d_{bb}^4 - 192d_{aa}^2d_{bb}^3d_{cc} + 576d_{aa}^2d_{bb}^2d_{cc}^2 - 192d_{aa}^2d_{bb}d_{cc}^3 + 384d_{aa}^2d_{cc}^4 - 192d_{aa}d_{bb}^5 + 192d_{aa}d_{bb}^4d_{cc} - 192d_{aa}d_{bb}^3d_{cc}^2 - 192d_{aa}d_{bb}^2d_{cc}^3 + 192d_{aa}d_{bb}d_{cc}^4 - 192d_{aa}d_{cc}^5 + 64d_{bb}^6 - 192d_{bb}^5d_{cc} + 384d_{bb}^4d_{cc}^2 - 448d_{bb}^3d_{cc}^3 + 384d_{bb}^2d_{cc}^4 - 192d_{bb}d_{cc}^5 + 64d_{cc}^6$

$ReE_{6,3}$ (163 terms; see the full text file)

$ReE_{6,6}$ (298 terms; see the full text file)

$ReE_{6,9}$ (280 terms; see the full text file)

$ReE_{6,12}$ (190 terms; see the full text file)

$ReE_{6,15}$ $6d_{aa}d_{ab}^5 - 6d_{aa}d_{ab}^4d_{ac} - 24d_{aa}d_{ab}^4d_{bc} + 6d_{aa}d_{ab}^3d_{ac}^2 + 12d_{aa}d_{ab}^3d_{ac}d_{bc} + 42d_{aa}d_{ab}^3d_{bc}^2 + 6d_{aa}d_{ab}^2d_{ac}^3 - 36d_{aa}d_{ab}^2d_{ac}^2d_{bc} + 18d_{aa}d_{ab}^2d_{ac}d_{bc}^2 - 48d_{aa}d_{ab}^2d_{bc}^3 - 6d_{aa}d_{ab}d_{ac}^4 + 12d_{aa}d_{ab}d_{ac}^3d_{bc} + 18d_{aa}d_{ab}d_{ac}^2d_{bc}^2 - 24d_{aa}d_{ab}d_{ac}d_{bc}^3 + 30d_{aa}d_{ab}d_{bc}^4 + 6d_{aa}d_{ac}^5 - 24d_{aa}d_{ac}^4d_{bc} + 42d_{aa}d_{ac}^3d_{bc}^2 - 48d_{aa}d_{ac}^2d_{bc}^3 + 30d_{aa}d_{ac}d_{bc}^4 - 12d_{aa}d_{bc}^5 + 6d_{ab}^5d_{bb} - 12d_{ab}^5d_{cc} - 24d_{ab}^4d_{ac}d_{bb} + 30d_{ab}^4d_{ac}d_{cc} - 6d_{ab}^4d_{bb}d_{bc} + 30d_{ab}^4d_{bc}d_{cc} + 42d_{ab}^3d_{ac}^2d_{bb} - 48d_{ab}^3d_{ac}^2d_{cc} + 12d_{ab}^3d_{ac}d_{bb}d_{bc} - 24d_{ab}^3d_{ac}d_{bc}d_{cc} + 6d_{ab}^3d_{bb}d_{bc}^2 - 48d_{ab}^3d_{bc}^2d_{cc} - 48d_{ab}^2d_{ac}^3d_{bb} + 42d_{ab}^2d_{ac}^3d_{cc} + 18d_{ab}^2d_{ac}^2d_{bb}d_{bc} + 18d_{ab}^2d_{ac}^2d_{bc}d_{cc} - 36d_{ab}^2d_{ac}d_{bb}d_{bc}^2 + 18d_{ab}^2d_{ac}d_{bc}^2d_{cc} + 6d_{ab}^2d_{bb}d_{bc}^3 + 42d_{ab}^2d_{bc}^3d_{cc} + 30d_{ab}d_{ac}^4d_{bb} - 24d_{ab}d_{ac}^4d_{cc} - 24d_{ab}d_{ac}^3d_{bb}d_{bc} + 12d_{ab}d_{ac}^3d_{bc}d_{cc} + 18d_{ab}d_{ac}^2d_{bb}d_{bc}^2 - 36d_{ab}d_{ac}^2d_{bc}^2d_{cc} + 12d_{ab}d_{ac}d_{bb}d_{bc}^3 + 12d_{ab}d_{ac}d_{bc}^3d_{cc} - 6d_{ab}d_{bb}d_{bc}^4 - 24d_{ab}d_{bc}^4d_{cc} - 12d_{ac}^5d_{bb} + 6d_{ac}^5d_{cc} + 30d_{ac}^4d_{bb}d_{bc} - 6d_{ac}^4d_{bc}d_{cc} - 48d_{ac}^3d_{bb}d_{bc}^2 + 6d_{ac}^3d_{bc}^2d_{cc} + 42d_{ac}^2d_{bb}d_{bc}^3 + 6d_{ac}^2d_{bc}^3d_{cc} - 24d_{ac}d_{bb}d_{bc}^4 - 6d_{ac}d_{bc}^4d_{cc} + 6d_{bb}d_{bc}^5 + 6d_{bc}^5d_{cc}$

$ReE_{6,18}$ $d_{ab}^6 - 3d_{ab}^5d_{ac} - 3d_{ab}^5d_{bc} + 6d_{ab}^4d_{ac}^2 + 3d_{ab}^4d_{ac}d_{bc} + 6d_{ab}^4d_{bc}^2 - 7d_{ab}^3d_{ac}^3 - 3d_{ab}^3d_{ac}^2d_{bc} - 3d_{ab}^3d_{ac}d_{bc}^2 - 7d_{ab}^3d_{bc}^3 + 6d_{ab}^2d_{ac}^4 - 3d_{ab}^2d_{ac}^3d_{bc} + 9d_{ab}^2d_{ac}^2d_{bc}^2 - 3d_{ab}^2d_{ac}d_{bc}^3 + 6d_{ab}^2d_{bc}^4 - 3d_{ab}d_{ac}^5 + 3d_{ab}d_{ac}^4d_{bc} - 3d_{ab}d_{ac}^3d_{bc}^2 - 3d_{ab}d_{ac}^2d_{bc}^3 + 3d_{ab}d_{ac}d_{bc}^4 - 3d_{ab}d_{bc}^5 + d_{ac}^6 - 3d_{ac}^5d_{bc} + 6d_{ac}^4d_{bc}^2 - 7d_{ac}^3d_{bc}^3 + 6d_{ac}^2d_{bc}^4 - 3d_{ac}d_{bc}^5 + d_{bc}^6$

# 6 $S(2) \times S(1)$ acting on $(a, b, p)$, $k = 1$

$|G| = 2$, $m = 9$ slots; $h = (ab)$ (order $r = 2$), $s = e$, mode scalar; real coefficients. Outputs: 11 (plain Bursarial: 26; quotient dimension 3).

Invariants output directly:

- inv[0] $= d_{aa} + d_{bb}$
- inv[1] $= 2d_{pp}$
- inv[2] $= 2d_{ab}$
- inv[3] $= d_{ap} + d_{bp}$
- inv[4] $= e_a + e_b$
- inv[5] $= 2e_p$

| $j$ | weight | real part of $z_j$ | imaginary part of $z_j$ |
|---|---|---|---|
| 0 | 1 | $2d_{aa} - 2d_{bb}$ | 0 |
| 1 | 1 | $2d_{ap} - 2d_{bp}$ | 0 |
| 2 | 1 | $2e_a - 2e_b$ | 0 |

| output | expanded |
|---|---|
| $inv[0]$ | $d_{aa} + d_{bb}$ |
| $inv[1]$ | $2d_{pp}$ |
| $inv[2]$ | $2d_{ab}$ |
| $inv[3]$ | $d_{ap} + d_{bp}$ |
| $inv[4]$ | $e_a + e_b$ |
| $inv[5]$ | $2e_p$ |
| $ReE_{2,0}$ | $-4d_{aa}^2 + 8d_{aa}d_{bb} - 4d_{bb}^2$ |
| $ReE_{2,1}$ | $-8d_{aa}d_{ap} + 8d_{aa}d_{bp} + 8d_{ap}d_{bb} - 8d_{bb}d_{bp}$ |
| $ReE_{2,2}$ | $-8d_{aa}e_a + 8d_{aa}e_b - 4d_{ap}^2 + 8d_{ap}d_{bp} + 8d_{bb}e_a - 8d_{bb}e_b - 4d_{bp}^2$ |
| $ReE_{2,3}$ | $-8d_{ap}e_a + 8d_{ap}e_b + 8d_{bp}e_a - 8d_{bp}e_b$ |
| $ReE_{2,4}$ | $-4e_a^2 + 8e_ae_b - 4e_b^2$ |